IS 2071-1 (1974): High voltage test techniques, Part 2: Test Procedure [ETD 19: High Voltage Engineering]

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Indian Standard

METHODS OF HIGH VOLTAGE TESTING

PART II TEST PROCEDURE

(*First Revision*)

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Gr 7 *September* 1974

(*Page* 24, *clause* **A-1.2.1,** *line* 4) — Substitute '$p = 0.5$' *for* '$p = 50.$'.

(*Page* 24, *clauses* **A-1.2.1** *and* **A-1.2.2**) — Substitute 'U' *for* 'U' wherever it appears.

(*Page* 24, *clause* **A-1.3**) — Substitute the following for the existing clause:

" **A-1.3 Confidence Limits** — From any set of n measurements, statistical checks may be made to define limits between which the true mean value U_m and the true standard deviation σ may be stated to lie with a given probability p_c of this being correct.

These limits are commonly expressed for $p_c = 0.95$ and are then termed '95 percent confidence limits'.

For Class 2 test results, and on the assumption that these have an approximately Gaussian distribution, the limits are given by the confidence limits for an arithmetic mean value. These are:

$$\overline{U} - s\, t_p / \sqrt{n} \leqslant U_m \leqslant \overline{U} + s\, t_p / \sqrt{n}$$

$$s\sqrt{(n-1) / x^2 p/2} \leqslant \sigma \leqslant s\sqrt{(n-1) / x^2 (1 - p^2)}$$

where $\overline{U}$ is the arithmetic mean of n voltage values, s is the standard deviation of n voltage values, t_p, $x^2\, p/2$ and $x^2\,(1 - p/2)$ are variable in the student's (t_p) and in the chi-square (x^2) distribution for $n-1$ degrees of freedom and with $p = 1 - p_c$. If no standard statistical table is available, the following values may be used for $p_c = 0.95$.

n	$t_p/\sqrt{n}$	$\sqrt{(n-1)/x^2 p/2}$ (Lower limit)	$\sqrt{(n-1)/x^2(1-p/2)}$ (Upper limit)
5	1·24	0·60	2·87
10	0·72	0·69	1·83
15	0·55	0·73	1·58
20	0·47	0·76	1·46
30	0·37	0·80	1·34
40	0·32	0·82	1·28
50	0·28	0·84	1·25

NOTE — That the confidence limits of σ are asymetrical.

For $n = 20$, the 95 percent confidence limits thus are given by:

$$\overline{U} - 0.47s \leqslant U_m \leqslant \overline{U} + 0.47\, s$$

$$0.76\, s \leqslant \sigma \leqslant 1.46\, s$$

In the case of Class 1 or Class 3 tests, other methods for the calculation of the confidence limits shall be used, for which reference should be made to the literature."

(*Page* 25, *clause* **A-1.4**) — Substitute the following for the existing clause:

'**A-1.4 Determination of Voltages Corresponding to Very Low or Very High Disruptive Discharge Probabilities** — For some purposes, a determination is desired of voltage levels corresponding to very low or very high probabilities of disruptive discharge of the test object.

Conventional rated withstand tests or assured disruptive discharge tests as described in **4.3.2** and **4.3.3** are neither intended, nor suitable for providing such information. The fact that a test object has passed such test procedures in itself gives little information concerning actual disruptive discharge probability.

Analysis of Class 1 test results as described in **A-1.2.1** gives more information on the probability distribution, but this is still not adequate for determining the test voltage levels giving defined low or high disruptive discharge probabilities. A modified test procedure is therefore described below for determining the voltage level corresponding to a very low disruptive discharge probability. (For high probability, *see* Note below).

As a preliminary test, several sets of three voltage applications are made, starting at a level U_0 below the estimated withstand voltage. After every set, the voltage level is increased by a constant amount Σ of between 2 percent and 5 percent of U_0. The series of tests are finished at the level U_1 when the first disruptive discharge occurs.

The test procedures are then continued in a similar manner, but with 25 voltage applications at each voltage level and starting at the level $U_2 = U_1 - 3\ \Sigma$. If no disruptive discharge occurs, the voltage level is increased after each series of 25 by successive steps of the same amount Σ, until a disruptive discharge occurs. When a disruptive discharge does occur, say during the series of applications at level U_3, the voltage is then reduced in successive steps equal to $2\ \Sigma$ until a full series of 25 applications has been applied with no disruptive discharge. Thereafter, the level is increased in steps equal to Σ until one more disruptive discharge is obtained.

The test result is then taken as the highest voltage value which has not given disruptive discharge during any series of 25 voltage applications. This value corresponds to a disruptive discharge probability of about 1 percent and there is about 98 percent certainty that it will be less than 8 percent. Usually, the complete test requires about 75 voltage applications.

NOTE — Almost identical procedures are used to determine the voltage corresponding to very high probabilities; the foregoing description applies if the words 'withstand' and 'probability of withstand' are replaced by 'disruptive discharge' and 'probability of disruptive discharge'. The voltage is first decreased from a high value instead of being increased from a low value.

Addenda

(*Page* 2, *clause* **0.4.1**, *line* 2) — Add the word 'test' after 'switching impulse voltage'.

(*Page* 24, *clause* **A-1.2**) — Add the following matter after the first paragraph:

> '*Class* 3 — Class 3 comprises tests made by repeated application of voltages of substantially constant shape in which the level, for each voltage applied, is determined by the result of the preceding voltage application, the first voltage applied being roughly equal to the estimated 50 percent disruptive discharge value.'

(*Page* 24, *clause* **A-1.2.2**) — Add the following new clause after **A-1.2.2**:

'**A-1.2.3** *Analysis of Results from Class 3 Tests* — A test procedure for making Class 3 tests and the method of analysis of test results so obtained to determine the 50 percent disruptive discharge voltage are described in **4.3.4** (b). For other methods and applications reference should be made to the literature.'

(*Page* 26, *clause* **A-1.4**) — Add the following new clause after **A-1.4**:

'**A-1.5 Conventional Rated Withstand and Assured Disruptive Discharge Tests** — The probability P of passing a conventional rated withstand test (*see* **4.3.2.2**) is given by the binominal distribution:

$$P = (1-p)^{15} + 15p(1-p)^{14} + \frac{15 \cdot 14}{2} p^2 (1-p)^{13}$$

where p is the probability of a disruptive discharge application at the test voltage.

> Using the test procedures described in **4.3.2**, therefore, and for a test voltage corresponding to $p = 0{\cdot}10$, the object will have 0·82 probability p of passing the test. With $p = 0{\cdot}01$, the corresponding value is $p = 0{\cdot}999\ 5$.
>
> Similarly, an assured disruptive discharge voltage test, according to **4.3.3** gives the probability Q of passing the test, where $Q = p^5 + 5(1-p)p^{14}$
>
> With $p = 0{\cdot}9$, the object will have 0·70 probability of passing the test and with $p = 0{\cdot}99$, the corresponding value is $Q = 0{\cdot}994$.

(ETDC 19)

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Indian Standard
METHODS OF HIGH VOLTAGE TESTING
PART II TEST PROCEDURES
(First Revision)

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0. FOREWORD

0.1 This Indian Standard (Part II) (First Revision) was adopted by the Indian Standards Institution on 17 January 1974, after the draft finalized by the High Voltage Techniques Sectional Committee had been approved by the Electrotechnical Division Council.

0.2 With a large number of high voltage and extra high voltage transmission system being constructed in this country, expensive electrical equipments are being put in service. It is necessary to ensure that such equipments are capable of withstanding the overvoltages normally encountered in service. This standard is intended to provide uniform methods of high voltage testing of electrical equipments.

0.3 This standard covers various aspects of high voltage testing, such as definition of terms, test procedures, requirements of the test objects and approved measuring devices for dielectric tests with direct, alternating, and impulse voltages, as well as impulse current. The Sectional Committee decided to withraw IS : 2070-1962 which covered impulse voltage testing.

0.4 The revision of this standard was undertaken in view of the advancements in high voltage technology.

0.4.1 This standard in addition, includes the procedures for carrying out switching impulse voltage, artificial pollution test and the statistical method of evaluating the test results.

0.5 This part (Part II) of the standard covers test procedures. The other two parts are:

Part I General definitions and requirements

Part III Approved measuring devices.

0.6 In the preparation of this standard, assistance has been derived from IEC Documents 42(Central Office)15, 20 and 20A Draft High-Voltage Test Techniques, Test Procedures issued by the International Electrotechnical Commission.

0.7 In reporting the result of a test made in accordance with this standard the final value, observed or calculated, is to be rounded off in accordance with IS : 2-1960*.

1. SCOPE

1.1 This standard (Part II) covers the test procedures and methods for generation and measurement of test voltage and currents for dielectric test with direct voltages, alternating voltages, impulse voltages and impulse currents. It also describes methods for the evaluation of test results and specifies the criteria for acceptance.

2. TEST WITH DIRECT VOLTAGE

2.1 Definitions

2.1.1 *Ripple* — Ripple is the periodic deviation from the arithmetic mean value. The magnitude of the ripple is defined as half the difference between the maximum and minimum values. The ripple factor is the ratio of the ripple magnitude to the arithmetic mean value of the voltage.

2.1.2 *Value of Test Voltage* — The value of the test voltage is defined by its arithmetic mean value.

NOTE — The maximum value of the voltage is approximately equal to the sum of the mean value and the ripple amplitude.

2.2 Test Voltage

2.2.1 *Requirements*

2.2.1.1 The test voltage with the test object connected should be a direct voltage with not more than a 5-percent ripple factor, unless otherwise specified in the relevant standards. Note that the presence of the test object may affect the ripple factor.

2.2.2 *Generation of Test Voltage*

2.2.2.1 The test voltage is generally obtained by means of rectifiers, or by electrostatic generators. The requirements to be met by the test voltage source depend considerably upon the type of the apparatus to be tested, and the test conditions. The requirements are determined mainly by the values and the nature of the test current to be supplied.

The important constituents of the test current are indicated in **2.2.4**. The output current rating of the source and the magnitude of the storage capacitance should be sufficient to charge the capacitance of the test object

*Rules for rounding off numerical values (*revised*).

in a reasonably short time. (This can take several minutes in some cases.) The source should be adequate to supply the leakage and absorption currents, and any internal and external partial discharge currents without excessive voltage drop which might influence the test results.

In tests on internal insulation these currents are usually small, but when testing wet insulators, leakage currents of the order of some tens of milliamperes or predischarge pulses of the order 10^{-3} coulomb may occasionally be encountered.

For test on polluted insulators, special circuit giving larger output currents are necessary. Provisionally it is recommended that the output impedance of the source should be such that when a pulse current of 500 mA is drawn for a duration of 500 ms the voltage drop during each pulse or a sequence of such pulses should not exceed 5 percent, the repetition rate of the current pulses being one per second.

2.2.3 *Measurement of Test Voltage*

2.2.3.1 *Measurement with approved devices* — The measurement of the arithmetic mean value, the maximum value or the ripple factor or any transient drop of the test voltage should be made with a device specified in Part III of this standard. Attention is drawn to the requirements on response characteristics when measuring ripples and transients.

2.2.3.2 *Measurement with sphere-gap* — The measurement of the maximum value of the voltage only can be made with a sphere-gap.

The procedure usually consists in establishing a relationship between the spacing at which disruptive discharge occurs and some other circuit variable related to the test voltage. (The relationship between the supply voltage and the output voltage may not be sufficiently stable for measuring purposes.) In establishing this relationship, the procedure specified in IS : 1876-1961* should be followed.

If this relationship is dependent on the presence of the test object, the sphere-gap, the precipitation in wet test, etc, it is important that the conditions should be the same during the calibration and the actual test.

The calibration is preferably made at 100 percent of the test voltage, but if necessary, extrapolation may be made from a voltage not lower than 80 percent of the test voltage. Attention is drawn to the possibility of erratic results due to pollution in the atmosphere.

2.2.4 *Measurement of Test Current*

2.2.4.1 When measurements of current through the test object are made, several separate components may be recognized which differ from each

*Method for voltage measurement by means of sphere-gaps (one sphere earthed).

other by several orders of magnitude for the same test object and test voltage as follows:

a) The capacitance current, due to the initial application of the test voltage and to any ripple or other fluctuations imposed on it;

b) The dielectric absorption current, due to slow charge displacements within the insulation and persisting for periods of a few seconds up to several hours. This process is reversible, currents of opposite polarity being observed when the test object is discharged and short-circuited;

c) The continuous leakage current, which is the final steady current attained at constant voltage after the components mentioned above have decayed to zero; and

d) Partial discharge currents.

2.2.4.2 Measurement of the first three components necessitates the use of instruments covering a wide range of magnitudes. From the observed current variation with time, information may sometimes be obtained concerning the condition of the insulation in non-destructive tests. Measurements of partial discharge pulse currents are made with special instruments which are dealt with in the IS : 6209-1971*.

2.2.4.3 The relative importance of each component depends on the type and the condition of the apparatus being tested, the purpose for which the test is made and the duration of the test. Accordingly, the measurement procedures should be specified in the relevant standard when it is required to distinguish a particular component.

2.2.5 *Test Procedure*

2.2.5.1 The voltage should be applied to the test object starting with a value sufficiently low to prevent overvoltage due to switching transients. It should be raised sufficiently slowly to permit accurate reading of the instruments but not so slowly as to cause unnecessary prolongation of the stressing of the test object near to the test voltage. These requirements are in general met if the rate of rise above 75 percent of the estimated final voltage is about 2 percent of this voltage per second.

2.2.5.2 The polarity of the voltage or the order in which the voltage of each polarity is applied, and also the rate of rise should be specified in the relevant standard, in those cases where deviations are required from the above specification.

2.2.5.3 *Assured withstand voltage test* — The test voltage should be raised to the specified value as described in 2.2.5.1. It should be maintained for the specified time and then reduced by discharging the smoothing capacitor through a suitable resistor. The test is generally satisfactory if no disruptive discharge occurs on the test object.

*Methods for partial discharge measurements.

2.2.5.4 *Assured disruptive discharge voltage test* — The test voltage should be raised as described in 2.2.5.1 until a disruptive discharge occurs on the test object. The value of the test voltage reached just prior to the disruptive discharge is the assured disruptive discharge voltage.

3. TEST WITH ALTERNATING VOLTAGES

3.1 Definitions

3.1.1 *Peak Value* — The peak value is the maximum value except that small high frequency oscillations arising for instance from partial discharges are disregarded.

3.1.2 *rms Value* — The rms value of an alternating voltage is the square root of the mean of the square of the voltage during a complete cycle.

3.1.3 *Value of Test Voltage* — The value of test voltage is defined by its peak value divided by $\sqrt{2}$.

NOTE — The relevant equipment standard may require a measurement of the rms value of the test voltage instead of the peak value for the cases where the rms value may be of importance. Such cases are for instance when thermal effects are under investigation.

3.2 Test Voltage

3.2.1 *Requirements*

3.2.1.1 The test voltage should be an alternating voltage having a frequency in the range 40-62 Hz or of an agreed value. Its form shall approximate to a sinusoid with both the half cycles closely alike and having a ratio of peak/rms values equal to $\sqrt{2}$ within ± 5 percent. The results of a high voltage test are considered unaffected by deviations from a sinusoid if these requirements are met.

3.2.1.2 There is insufficient information available to indicate whether these requirements are generally met by tests circuits in common use; hence somewhat greater deviation from a sinusoid may have to be accepted. Note that the presence of the text object, especially those having non-linear impedance characteristics, may considerably effect the deviation from a sinusoid.

NOTE — A useful measure, for the specification or acceptance testing of test equipment, is the rms value of the harmonics. It can generally be assumed that the above requirements on deviations from a sinusoid will be met if this value does not exceed 5 percent of the rms value of the fundamental.

3.2.2 *Generation*

3.2.2.1 The test voltage is generally supplied from a step-up transformer but alternatively, it may be generated by means of a series resonant circuit.

NOTE — Attention is drawn to the possibility that non-sustained external disruptive discharges may cause a large overswing between the terminals of the test object. The phenomenon may cause failure or flashover of the test object or of the testing transformer. A cure can usually be effected by changing the natural frequency of the voltage source, or by introducing some attenuation into the system.

3.2.2.2 *Test transformer* — The voltage in the test circuit should be stable enough to be practically unaffected by varying leakage currents. Partial discharges or predischarges in the test object should not reduce the test voltage to such an extent and for such a time that the measured disruptive discharge voltage of the test object is affected. This is usually achieved if:

a) the total capacitance of the test object and any additional capacitance is not less than about 1 000 pF.

NOTE — If any external protective series resistance does not exceed 1 kΩ, the transformer effective terminal capacitance may be regarded as being in parallel with the test object.

b) the steady state current delivered by the transformer, when the test object is short-circuited at the test voltage, is not less than 1A rms. Exceptions are:

   i) Tests on small samples of solid insulation, insulating liquids, or combination of the two. For such tests, a short-circuit current of the order of 0·1A rms may suffice.

   ii) Tests under artificial pollution. For such tests, the requirements of the following table should be met where R_s is the total series resistance and X_s the steady-state series reactance at the test frequency of the supply circuit including the generator or the network.

R_s/X_s	*Short-Circuit Current I,* A rms
≤0·1	6
0·1-0·3	12

3.2.2.3 *The series resonant circuit* — The series resonant circuit consists essentially of a capacitive test object or load connected in series with a variable inductance to a medium voltage power supply. By varying the inductance the circuit can be tuned to resonance at the required frequency under which condition a voltage of a shape approximating very closely to a sinusoid is applied to the test object. The stability of the resonance conditions and of the voltage depend on the constancy of the test circuit impedances.

The method is especially useful when testing objects such as capacitors and cable samples, when the external insulation effects are swamped by the capacitance of the load. The circuit is unsuitable for tests when the resonant conditions are seriously disturbed by non-linear leakage currents or discharges, for example for tests on external insulation under wet and polluted conditions.

3.2.3 *Measurement*

3.2.3.1 *Measurement with approved devices* — The test voltage and the deviation from a sinusoid should be measured with devices specified in Part III of this standard.

3.2.3.2 *Measurement with sphere-gap* — The measurement of the peak value of the test voltage only may be made with a sphere-gap. The procedure usually consists in establishing a relationship between the spacing at which disruptive discharge occurs and some other circuit variable related to the test voltage. In establishing this relationship, the procedure specified in IS : 1876-1961* should be followed.

If this relationship is dependent on the presence of the test object, the sphere-gap, the precipitation in wet tests, etc, it is important that the conditions should be the same during the calibration and the actual test.

The calibration is preferably made at 100 percent of the test voltage, but extrapolation may be made from a voltage not lower than 80 percent of the test voltage. Extrapolation may be unsatisfactory if the current of the test circuit is not proportional to the voltage, or if modifications have occurred in the voltage shape or in the frequency at the 100 percent voltage level.

3.2.4 *Test Procedure*

3.2.4.1 The voltage should be applied to the test object starting at a value sufficiently low to prevent overvoltages due to switching transients. It should be raised sufficiently slowly to permit accurate reading of the measuring instrument, but not so slowly as to cause unnecessary prolongation of the stress near the test voltage.

These requirements are in general met if the rate of rise above 75 percent of the estimated final test voltage is about 2 percent per second of this voltage. However, certain types of objects may necessitate a different rate of rise which should be specified in the relevant standards.

3.2.4.2 *Assured withstand voltage test* — The test voltage should be raised to the specified value as described in **3.2.4.1**. It should be maintained for the specified time and then rapidly decreased. The voltage should not be interrupted suddenly to avoid the possibility of switching transients which may affect the subsequent test results. The test is satisfactory if no disruptive discharge occurs on the test object, unless other acceptance criteria are specified in the relevant equipment specifications.

3.2.4.3 *Assured disruptive discharge voltage test* — The test voltage should be raised as described in **3.2.4.1** until a disruptive discharge occurs on the test object. The value of the test voltage reached just prior to the disruptive discharge is the assured disruptive discharge voltage.

4. TEST WITH LIGHTNING IMPULSE VOLTAGES

4.1 Definitions

4.1.1 *Full Lightning Impulse* — A full lightning impulse (*see* Part I of this standard) is a lightning impulse which is not interrupted by a disruptive discharge.

*Method for voltage measurement by means of sphere-gaps (one sphere earthed).

4.1.2 *Chopped Lightning Impulse* — A chopped lightning impulse is a lightning impulse which is interrupted by the disruptive discharge of an external chopping gap or of the external insulation of the test object, causing a sudden collapse of the voltage, practically to zero value. The collapse can occur on the wave front, at the peak or on the tail.

4.1.3 *Value of Test Voltage* — The value of the test voltage is normally the peak value. With some test circuits oscillations or overshoot may be present on the voltage. If the amplitude of the oscillations is not greater than 5 percent of the peak value and the frequency is at least 0·5 MHz, or alternatively, if the amplitude of the overshoot is not greater than 5 percent of the peak value and the duration not greater than 1 μs, a mean curve may be drawn for the purpose of measurement, the maximum amplitude of which is defined as the value of the test voltage (*see* Fig. 1A). For other cases (*see*, for instance, Fig. 1B), the relevant standards should define the value of the test voltage taking into account the type of test and test object.

NOTE — In what follows, the term 'peak value' includes the term 'virtual peak value' unless otherwise stated.

4.1.4 *Virtual Front Time T_1* — The virtual front time T_1 is defined as 1·67 times the time interval T between the instants when the impulse is 30 percent and 90 percent of the peak value (*see* points A, B in Fig. 2 to 4). If oscillations are present on the wave front, the points A and B should be taken on the mean curve drawn through these oscillations.

4.1.5 *Virtual Origin O_1 of an Impulse* — The virtual or gin O_1 of an impulse is defined as the instant preceding that corresponding to the point A (*see* Fig. 2 to 4) by a time 0·3 T_1. For oscillograms having linear time sweeps, this is the intersection with x-axis of a straight line drawn through the reference point A and B on the wave front.

4.1.6 *Virtual Time to Half Value T_2* — The virtual time to half value T_2 of an impulse is the time interval between the virtual origin and the instant on the tail, when the voltage has decreased to half of the peak value.

4.1.7 *Definitions Applicable Only to Chopped Impulses* — Ideally the chopping of an impulse is characterised by an initial discontinuity which decreases the voltage. The voltage then continues to decrease to zero or nearly zero, either directly or via oscillations.

NOTE — With some impulse measuring systems, test objects or test arrangements, oscillograms of chopped impulses show a rounding off of the voltage or a step which may be followed by a partial recovery, or even may cause a double peak before the final collapse of the voltage. Such cases are not dealt with in this standard.

4.1.7.1 *Virtual time of voltage collapse during chopping* — The virtual time of voltage collapse during chopping is 1·67 times the time interval between points C and D (*see* Fig. 3 and 4).

4.1.7.2 *Instant of chopping* — The instant of chopping is the instant when the initial discontinuity occurs.

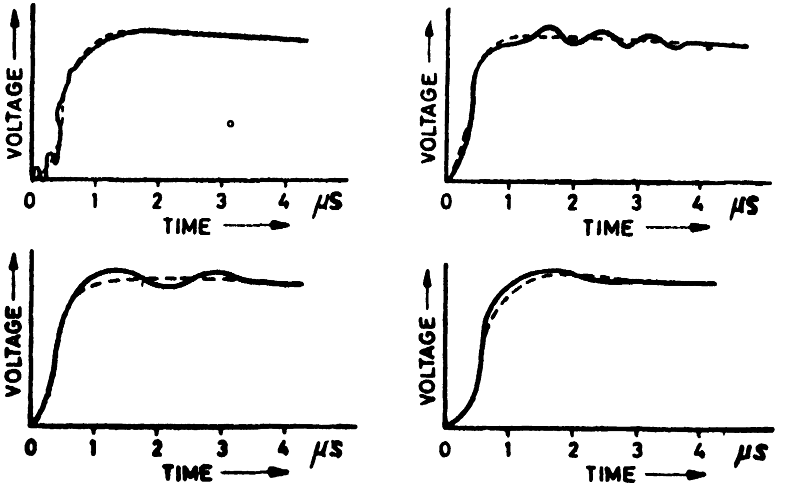

Fig. 1A Construction for Derivation of Virtual Peak Values

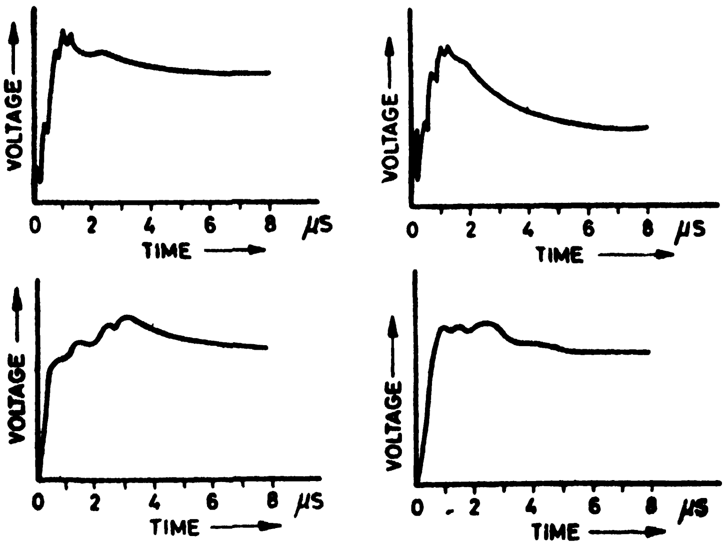

Fig. 1B Irregular Impulse Shapes

4.1.7.3 *Virtual time T_c to chopping* — The virtual time T_c to chopping is the time interval between the virtual origin O_1, and the virtual instant of chopping.

4.1.7.4 *Virtual steepness of voltage during chopping* — The virtual steepness of voltage during chopping is the quotient of the estimated voltage at the instant of chopping and the virtual time of voltage collapse.

NOTE — The use of points *C* and *D* are for definition purposes only; it is not implied that the steepness of chopping can be measured with any degree of accuracy using conventional measuring circuits.

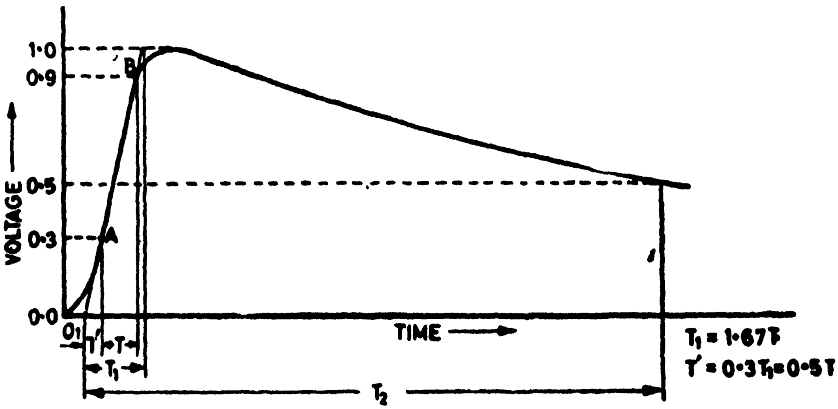

FIG. 2 FULL IMPULSE VOLTAGE

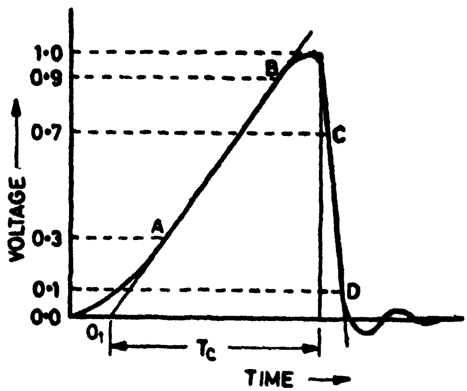

FIG. 3 IMPULSE VOLTAGE CHOPPED ON THE FRONT

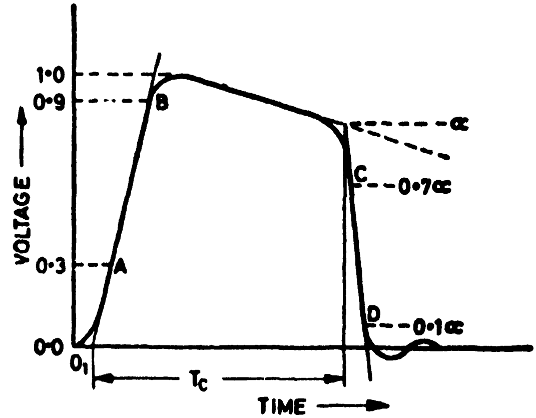

Fig. 4 Impulse Voltage Chopped on the Tail

4.1.8 *Linearly Rising Front-Chopped Impulses* — A voltage rising with approximately constant steepness, until it is chopped by the occurrence of a disruptive discharge, is described as a linearly rising impulse (*see* Fig. 5).

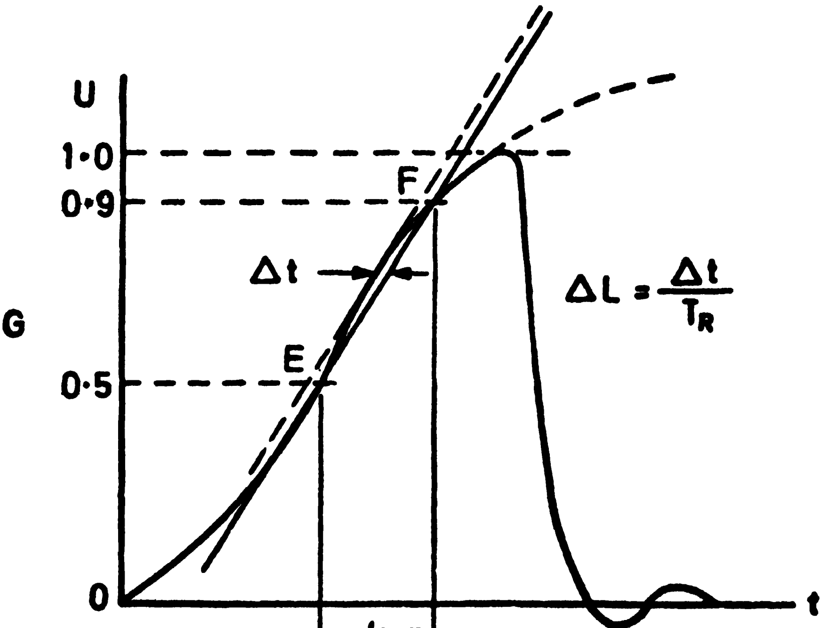

Fig. 5 Linearly Rising Impulse

The parameters of the impulse are determined by two points E and F, located on the wave front of the impulse or, if oscillations are present, on a mean curve through them at 50 percent and 90 percent of the peak value.

The impulse is defined by:

— the peak value at the instant of chopping as measured on the oscillogram.

— the rise time T_R. This is the time interval E-F (*see* Fig. 5) multiplied by 2·5.

— The virtual steepness, S. This is the slope of the straight line E-F, usually expressed in KV/µs.

To judge whether the wave front can be considered approximately linear, two straight lines are drawn parallel to the line E-F such that they enclose the entire impulse from 50 percent up to the instant of chopping.

If the time difference corresponding to these two lines is not more than 0·05 times the rise time, the impulse can be considered approximately linear (*see* Note).

NOTE — To avoid excessive deviation from linearity it is recommended that the prospective peak value of the impulse should not be less than 1·1 times the peak value at the instant of chopping.

4.1.9 *Curves Relating the Disruptive Discharge Voltage Peak to the Times to the Disruptive Discharge (Voltage/Time Curves)*

4.1.9.1 *Voltage/time curve for wave fronts rising linearly* — The voltage/time curve for impulses with wave fronts rising linearly is the curve relating the disruptive discharge voltage of a test object to T_R.

The curve is obtained by applying impulses with approximately linear wave fronts of different steepnesses.

4.1.9.2 *Voltage/time curve for impulses of constant prospective shape* — The voltage/time curve for impulses of constant prospective shape is the curve relating the disruptive discharge voltage of a test object to the virtual time to chopping which may occur on the wave front, at the peak or on the tail. The curve is usually obtained by applying impulse voltages of constant shape but with different peak values (*see* Fig. 6).

4.2 Test Voltage

4.2.1 *Standard Lightning Impulse* — The standard lightning impulse is a full lightning impulse having a virtual front time of 1·2 µs and a virtual time to half value of 50 µs. It is described as a 1·2/50 impulse.

4.2.2 *Tolerances* — If not otherwise specified in the relevant standards, the following differences between specified values for the standard impulse and those actually recorded are permitted when the measuring system meets the requirements of Part III of this standard.

Peak value	± 3 percent
Front time	± 30 percent
Time to half value	± 20 percent

Overshoot or oscillations in the neighbourhood of the peak value are tolerated provided their single peak amplitude is not larger than 5 percent of the peak value. Oscillations on the first part of the impulse (below 50 percent of the peak value) are tolerated provided their single peak amplitude does not exceed 25 percent of the peak value.

NOTE 1 — In specific cases, such as during tests on low impedance objects, it may be difficult to adjust the shape of the impulse within the tolerances recommended, to keep the oscillations or overshoot within the specified limits, or to avoid a polarity reversal. Such cases shall be dealt with separately in the relevant equipment standard.

NOTE 2 — It is emphasized that the tolerances mentioned above constitute the permitted differences between the specified values and those actually recorded by measurements. These differences should be distinguished from measuring errors which are the difference between the true values and those actually recorded.

No corrections are accepted for measuring errors. Note however that the measurement of impulses chopped at the front at times shorter than 0·5 μs, especially those of high voltage, presents considerable technical difficulties. The same applies to the measurement of oscillations on the wave front especially at frequencies above one MHz.

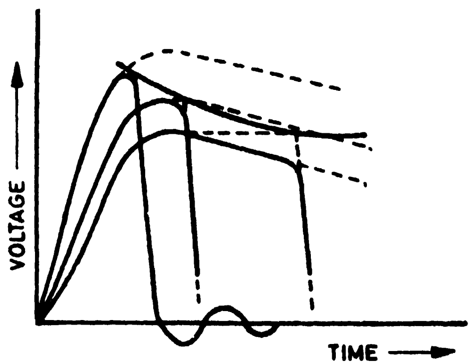

FIG. 6 VOLT/TIME CURVE FOR STANDARD IMPULSES

4.2.3 *Standard Chopped Lightning Impulse* — A standard chopped lightning impulse is a standard impulse which is chopped after 2-5 μs. Because of incomplete technical information, the virtual time of voltage collapse has not yet been standardized. This characteristic is of importance only for some specific tests.

4.2.4 *Generation and Requirements of Test Voltage* — The impulse is usually generated by an impulse generator which essentially is an apparatus in which a number of capacitors are charged in parallel from a direct voltage source and then discharged in series via a circuit which includes the test object. The values of the circuit parameters should be such that the required impulse shape is obtained with the test object in the circuit.

4.2.5 *Determination of Impulse Shape* — The impulse shape should be adjusted with the test object in circuit. For this purpose, an approved divider and oscillograph should be used (*see* Part III of this standard). The impulse shape should be measured in accordance with **4.1.4** to **4.1.8**.

In general, the impulse shape should be checked for each test object. Where a number of test objects of the same design and size are tested under unchanged conditions, the shape need only be verified once.

NOTE — A determination of the impulse shape by calculation is not satisfactory.

4.2.6 *Measurement of Test Voltage*

4.2.6.1 *Measurement with approved devices* — The test voltage should be measured by a device specified in Part III of this standard.

4.2.6.2 *Measurement with sphere-gap* — The measurement of the peak value of the test voltage may be made with a sphere-gap. The procedure usually consists in establishing a relationship between the spacing at which disruptive discharge occurs and some other variable in the circuit related to the test voltage, for instance the maximum charging voltage of the first stage of the impulse generator. In establishing this relationship, the procedure specified in IS : 1876-1961* should be followed. If this relation is dependent on the presence of the test object, the sphere-gap, etc, it is important that the conditions are the same during the calibration and the actual test except that the sphere-gap shall be opened sufficiently to prevent sparkover during the test.

For tests on insulation which is not likely to be damaged by repeated voltage applications or flashovers, calibration is made up to nearly 100 percent of the test voltage.

For tests on other objects, extrapolation may be unavoidable but such extrapolation should not be greater than 60 percent of the test voltage. The extrapolation is only permissible if it can be shown that the test voltage is proportional to the related quantity.

4.3 Test Procedure

4.3.1 The test procedure applicable to particular items of equipment, for example the polarity to be used, the preferred order if both polarities are

*Method for voltage measurement by means of sphere-gaps (one sphere earthed).

to be used, the number of applications, and the interval between applications, should be specified in the relevant equipment standard; having regard to such factors as:

a) the accuracy of test results,
b) the statistical nature of the observed phenomenon and the polarity dependence of the measured characteristic, and
c) the possibility of progressive deterioration with repeated voltage applications.

The test procedures recommended below are applicable to test objects which are not subject to progressive deterioration by repeated voltage applications. For other types of test objects acceptance criteria other than those specified in **4.3.2** to **4.3.5** may be needed.

4.3.2 *Rated Withstand Voltage Test*

4.3.2.1 *Tests on nonself-restoring insulation* — Unless otherwise specified for tests on nonself-restoring insulation, three impulses of the specified shape and polarity are applied at the rated withstand voltage level. The requirements of the test are satisfied if no indication of failure is obtained, using the methods of detection specified in the relevant equipment standard.

4.3.2.2 *Tests on self-restoring insulation and on combinations of self and nonself-restoring insulation* — A procedure usually recommended for test objects having combinations of self-restoring and nonself-restoring insulation is given below:

Fifteen impulses of the rated withstand voltage and of the specified shape and polarity are applied. The requirements of the test are satisfied if no disruptive discharge occurs in the nonself-restoring insulation and not more than two disruptive discharges occur in the self-restoring insulation.

In the cases of tests objects having self-restoring insulation only, the test object can also be deemed satisfactory if the 50 percent disruptive discharge voltage, determined according to **4.3.3** is not less than $1/(1 - 1{\cdot}3\sigma)$ times the rated impulse withstand voltage, where σ is the per unit standard deviation of the disruptive discharge voltage. Unless otherwise recommended in the relevant equipment standard the following values should be used for air insulation:

— Lightning impulse tests, $\sigma = 0{\cdot}03$

— Switching impulse tests, $\sigma = 0{\cdot}06$

NOTE — In case of doubt, the relevant equipment standard is to decide in which of the above categories a particular type of equipment should be placed.

4.3.3 *Assured Disruptive Discharge Voltage Test* — This test is intended as a check that a specified test voltage will cause a disruptive discharge with a given high probability (*see also* Appendix A). Five impulses of the specified prospective peak value, shape and polarity are applied. If each impulse causes a disruptive discharge, the requirements of the test have been satisfied. If more than one impulse fails to cause a disruptive discharge, the requirements of the test have not been satisfied. If one impulse fails to cause disruptive discharge, ten additional impulses are applied. If each of these additional impulses causes a disruptive discharge, the requirements of the test have been satisfied.

4.3.4 *Fifty Percent Disruptive Discharge Voltage Test* — The voltage corresponding to a 50 percent disruptive discharge probability U, 50 percent may be determined as follows (*see also* Appendix A).

a) At least ten impulses are applied at each of the several test voltage levels, the steps in voltage being 2-5 percent of the expected 50 percent disruptive discharge voltage. The value of the 50 percent disruptive discharge voltage is found from a curve of disruptive discharge probability versus prospective test voltage. The accuracy of the determination increases with increased number of voltage applications but there is usually no need to go above 20 applications at each level.

   The same method can be used to determine the standard deviation of the flashover voltage. See Appendix A, which also gives some information on confidence limits both for the 50 percent voltage and for the standard deviation.

b) The following is an alternative procedure to determine the 50 percent value. A voltage U_k is chosen which is considered approximately equal to the 50 percent disruptive discharge level. A voltage interval ΔU, approximately 3 percent of U_k, is also chosen. One impulse is applied at the level U_k. If this does not cause a disruptive discharge, the next impulse should have the level $U_k + \Delta U$. If a disruptive discharge occurs at the level U_k, the next impulse should have the level $U_k - \Delta U$.

   This procedure is repeated, a number of times each impulse having a level determined by the effect of the proceeding impulse. The number of impulse n_v at each voltage level U_v, is counted and the 50 percent disruptive discharge voltage is determined as:

$$U_{50} \text{ percent} = \frac{\Sigma n_v \quad U_v}{\Sigma n_v}$$

   In this equation, the first level taken into account should be one at which two or more impulses were applied. This partially corrects for the error which may be introduced if U_k is much too low or much too high. For normally distributed flashover probabilities, the indicated value of U_{50} percent determined as above for $\Sigma n_v - \geqslant 20$ will generally lie within the range from 30 percent to 70 percent of the true disruptive discharge probability.

4.3.5 *Determination of Very Low or Very High Disruptive Discharge Probabilities* — Some guidance for the determination of voltages corresponding to very low and very high disruptive discharge probabilities is found in Appendix A. These procedures are not intended for conventional acceptance testing but may be useful in other cases for study related to insulation coordination.

5. TESTS WITH SWITCHING IMPULSE VOLTAGE

5.1 Definitions

5.1.1 *Switching Impulse* — A switching impulse which should be distinguished from a lightning impulse is defined in Part I of this standard.

5.1.2 *Value of Test Voltage* — If not otherwise specified in the relevant standards, the value of the test voltage is the prospective peak value.

5.1.3 *Time to Peak* T_{cr} — The time to peak T_{cr} is the time interval between actual zero and the instant when the voltage has reached its peak value.

5.1.4 *Time to Half Value* T_h — The time to half value T_h is the time interval between actual zero and the instant on the tail when the impulse has first decreased to half the peak value.

5.1.5 *Time Above* 90 *Percent* T_d — The time above 90 percent T_d is the time interval during which the impulse voltage exceeds 90 percent of its peak value.

Note to clauses **5.1.3** *to* **5.1.5** — Practical reasons and theoretical considerations have caused several different time parameters to be used to characterize a switching impulse. Two such parameters related to the wave front are the time to peak and the virtual front time. The former is defined in **5.1.3**, the latter is defined either by analogy with virtual front time for lightning impulses (*see* **4.1.4**) or by using some different reference points and multiplying factors. The difference between wave front times defined according to these various measures is usually small. For switching impulses with time parameter as given in **5.2.1** and **5.2.2**, the time to peak is between 1·4-1·8 times the virtual wave front time.

Two other time parameters are related to the duration of the impulse; they are the time to half value and the time above 90 percent. The latter measurement is found practical in the cases where the wave form of the impulse is dictated, for instance, by the saturation phenomena in the test object or the test circuit, or when it is felt that the stress of the test object is highly dependent on this parameter as for some kind of internal insulation. These time parameters are defined in **5.1.4** and **5.1.5**. It should be emphasized that, when defining a switching impulse, only one parameter related to the wave front and one to the duration should be used.

5.1.6 *Time to Chopping T_c* — The time to chopping T_c of a switching impulse is the time interval between the actual zero and the moment when the chopping occurs.

5.1.7 *Linearly Rising Switching Impulse* — The linearly rising impulse is defined in 4.1.8.

5.2 Test Voltage

5.2.1 *Standard Switching Impulse* — The standard switching impulse is an impulse having a time to peak of 250 μs and a time to half value of 2 500 μs. It is described as a 250/2 500 impulse.

5.2.2 *Special Impulses* — In the cases where the standard impulse alone is not considered to be appropriate or adequate, impulses of 100/2 500 and 500/2 500 are recommended. Also an oscillatory impulse has been proposed, which has a duration of the first half period between 2 000 and 3 000 μs and the peak value of the second half period as high as possible up to about 80 percent of the peak value of the first one.

5.2.3 *Tolerances* — If not otherwise specified in the relevant equipment standards the following differences between recorded and specified values are accepted both for standard and special impulses when the measuring device meets the requirement of Part III of this standard.

Peak value	± 3 percent
Front time	± 20 percent
Time to half value	± 60 percent

NOTE 1 — In specific cases, such as during tests on low impedance objects, it may be difficult to adjust the shape of the impulse within the tolerances recommended, to keep the oscillations or overshoot within the specified limits, or to avoid a polarity reversal. Such cases shall be dealt with separately in the relevant equipment standard.

NOTE 2 — The measured results of tests using switching impulses with linear wave fronts are usually present as curves showing the times to disruptive discharge, or alternatively virtual steepness on one axis and disruptive discharge voltage on the other. For this reason, and because the range of practical interest is very large, there is no need to specify any particular wave front time or steepness to be used, nor any tolerances on these values.

5.2.4 *Generation and Requirements of Test Voltage* — Switching impulses are usually generated by an impulse generator (*see* 4.2.4). They can also be generated by discharging a capacitor through the primary winding of a testing transformer. The basic circuit arrangement for the latter method is shown in Fig. 7 where C_1 is the capacitor to be discharged and K is a switch. The circuit elements within the dotted area used for adjusting the impulse shape, R_2 and C_2 merely being shown as one possible arrangement for this purpose. Other methods are also in use involving current through a transformer winding.

NOTE — When testing a transformer with switching impulses, the transformer itself may be used for generating the impulse by the method described for testing a transformer.

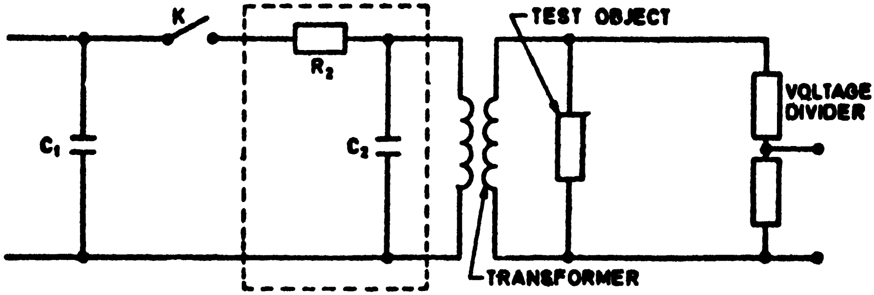

FIG. 7 GENERATION OF SWITCHING IMPULSES

The elements of a circuit for the generation of switching impulses should be so chosen as to avoid excessive distortion of the impulse shape due to pre-discharge currents in the test object. Such currents may reach considerable values especially during pollution tests on external insulation at high voltages. In test circuits of high internal impedance, distortion of the voltage due to such currents may prevent the occurrence of disruptive discharge.

5.2.5 *Measurement of Test Voltage* — The test voltage should be measured with a device specified in Part III of the standard. Alternatively, it may be measured with a sphere gap (*see* IS : 1876-1961*). However, it may be noted that this standard gives no information specially related to switching impulses.

5.2.6 *Test Procedure* — The test procedure is the same as for lightning impulse testing (*see* 4.3).

6. IMPULSE CURRENT

6.1 Definitions

6.1.1 *Impulse Currents* — Two types of impulse currents are dealt with. One has a shape which increases from zero to the peak value in a short time and thereafter decreases to zero approximately exponentially or in the manner of a heavily damped sine curve. This type is designated by the wave front time and the time to half value, *see* 6.1.3 and 6.1.7.

The other type has an approximately rectangular shape and is designated by the virtual duration of the peak and the total virtual duration (*see* 6.1.6 and 6.1.7).

6.1.2 *Value of Test Current* — The value of the test current is normally defined by the peak value. With some test circuits overshoot of oscillations

*Method for voltage measurement by means of sphere-gaps (one sphere earthed).

may be present on the current. It will depend on the type of the test and also whether the value of the test current shall be defined by the actual peak or a smooth curve drawn through these oscillations.

6.1.3 *Virtual Front Time* T_1 — The virtual front time T_1 is defined as 1·25 times the interval between the instants when the impulse current is 10 percent and 90 percent of the peak value.

If oscillations are present on the wave front, the 10 percent and 90 percent values should be derived from the mean curve drawn through these oscillations.

NOTE — The difference between wave front times measured according to this definition and the one in 4.1.4 is generally less than 10 percent.

6.1.4 *Virtual Origin* O_1 — The virtual origin O_1 of an impulse current is defined as the instant preceding that corresponding to the time at which the current is 10 percent of the peak value by 0·1 T_1.

For oscillograms having linear time sweeps this is equivalent to the intersection with the x-axis of a straight line drawn through the 10 percent and 90 percent points on the wave front.

6.1.5 *Virtual Time to Half Value,* T_2 — The virtual time to half value, T_2 is the time interval between the virtual origin and the instant on the tail, where current has first decreased to half the peak value.

6.1.6 *Virtual Duration of the Peak of a Rectangular Impulse Current* — The virtual duration of the peak of a rectangular impulse current is defined by the time during which the current is greater than 90 percent of the peak value.

6.1.7 *Virtual Total Duration of a Rectangular Impulse* — The virtual total duration of a rectangular impulse current is the time during which the amplitude of the impulse is greater than 10 percent of its peak value. If small oscillations are present on the wave front, a mean curve should be drawn to determine the time at which the 10 percent value is reached.

6.2 Test Current

6.2.1 *Standard Impulse Currents* — Two standard impulse currents are used. One has a virtual wave front time of 8 µs and a time to half value of 20 µs; it is described as an 8/20 impulse. The other has a virtual wave front time of 4 µs and a time to half value of 10 µs; it is described as 4/10 impulse. Rectangular impulse currents usually have virtual durations at the peak within 2 000 to 3 500 µs.

6.2.2 *Tolerances*

6.2.2.1 The following tolerances on the actual recorded values are permitted when the measuring system meets the requirements of Part III of this standard.

For 8/20 and 4/10 impulses

Peak value	± 10 percent
Virtual front time, T_1	± 10 percent
Virtual time to half value, T_2	± 10 percent

Small overshoot and oscillations are tolerated provided that their single peak amplitude in the neighbourhood of the peak value of the impulse is not more than 5 percent of the peak value. Any polarity reversal after the current has fallen to zero, should be not more than 20 percent of the peak value.

For rectangular impulses

Peak value	+ 20 percent − 0 percent
Virtual duration of the peak	+ 20 percent − 0 percent

Overshoot and oscillations are tolerated provided that their single peak amplitude is not more than 10 percent of the peak value. The total duration of a rectangular impulse should not be more than 1·5 times the virtual duration of the peak and the polarity reversal should be limited to 10 percent of the peak value.

6.2.3 *Measurement of Test Current* — The test current should be measured by a device specified in Part III of this standard.

6.2.4 *Measurement of Voltage During Tests with Impulse Currents*

6.2.4.1 It is sometimes required to measure the voltage developed across the test object during tests with high impulse currents. Anyone of the approved devices listed in Part III of this standard can be used for the purpose.

The impulse current may induce high voltages in the impulse voltage measuring circuit, thereby giving measuring errors. To check this, it is recommended that the conductor which normally joins the voltage divider to the live end of the test object, should be disconnected from the live end and connected to the earthed end of the test object but maintained in approximately the same loop. Alternatively, the test object is short-circuited or replaced by a metal conductor. The voltage measured under this condition when the generator is discharged should be substantially zero at least during the part of the impulse. which is of importance for evaluating the results.

NOTE — This short-circuit check may be made at a reduced current.

APPENDIX A

(*Clauses* 4.3.3, 4.3.4 *and* 4.3.5)

STATISTICAL PROCESSING OF THE TEST RESULTS

A-1. GENERAL

A-1.1 Statement of the Problem — As disruptive discharges are a random phenomenon, statistical procedures are important to obtain more precise information from the tests. It is assumed that, to each test voltage level U, a probability p is attached, for a disruptive discharge to take place at that level. The aim of the tests is to determine the probability distribution function p (U) relating p to U.

Usually the probability function is expressed in terms of certain parameters out of which the mean value U_m and the standard deviation σ are dealt with below. These parameters and the form of the function can be determined from a very large number of voltage applications if successive voltage applications do not change significantly the characteristics of the test object, or if a fresh test object is used for each application. In certain cases, such as impulse tests on *external insulation*, the probability distribution function is found to be approximately Gaussian from a few up to 95 or even 98 percent flashover probability. The standard deviation usually is between 2 and 8 percent. Very large values of the standard deviation have been found during impulse tests in internal insulation.

Little information is available on the character of the probability distribution function for internal insulation for any type of voltage and for external insulation for direct and alternating voltage.

In practical tests, the number of voltage applications is limited. Then the parameters are determined only within certain limits with a certainty which can be determined. The values so determined are called U and s.

A brief summary of some simple methods which have been found useful in the analysis of high voltage test results are given below. The methods are based on the above assumption of undisturbed test objects.

NOTE — Attention is drawn to the fact that the distribution function is not necessarily symmetrical with reference to the average. Thus, it is necessary to distinguish between the terms ' the average value ' and the ' the most probable value '.

A-1.2 Classification of Test Results — Test results can be subdivided into two categories to permit statistical evaluation.

Class 1 — This class comprises tests made by repeated application of voltages of substantially constant shape in which, for each voltage level, the proportion of voltage applications causing disruptive discharges is recorded. It mainly applies to impulse tests but certain alternating and direct voltage tests belong to this class.

Class 2 — This class comprises procedures in which each application of the test voltage causes a disruptive discharge. The tests are made by applying continuously increasing voltages to the test object, and by measuring the actual disruptive discharge voltages obtained. Such tests can be made with direct, alternating or impulse voltages. Tests causing disruptive discharge on the wave front of an impulse belong to this class.

A-1.2.1 *Analysis of Class 1 Test Results* — The results of Class 1 tests can be plotted on Gaussian linear paper (probability paper) with the voltage on the linear axis. If they lie approximately a straight line, then the distribution is approximately Gaussian. The voltage corresponding to $p = 50$ can be used as determination of the mean value U and the voltage range between $p = 0{\cdot}5$ and $p = 0{\cdot}16$ as an estimation of s.

Other more accurate methods to determine U and s are found in the literature.

A-1.2.2 *Analysis of Class 2 Test Results* — The results of Class 2 tests appear as a series on n voltage values $U\nu$ from which the mean flashover voltage U and the standard deviations can be obtained:

$$\bar{U} = \frac{1}{n}\sum_{1}^{n} U\nu$$

$$s^2 = \frac{1}{n-1}\sum_{1}^{n} (U\nu - \bar{U})^2$$

Alternatively, $n\nu/n+1$ can be plotted on a probability paper as a function of $U\nu$ where $n\nu$ is the number of flashovers up to and including the voltage $U\nu$ and n is the total number of voltage applications. The curve permits a determination of U and s in the same manner as in **A-1.2.1** but does not necessarily give the same result as the numerical method described above.

A-1.3 Confidence Limits — From any set of n measurements, statistical checks may be made to define the limits between which the true mean value U_m and the true standard deviation σ may be stated to be with a given probability p_e of being correct. These limits are commonly expressed for $P_e = 0{\cdot}95$ and are then termed ' 95 percent confidence limits '.

The limits for the true mean value U_m are then given by:

$$U_m = \bar{U} \pm t_p \frac{s}{n}$$

where t_p is a function of p_e and n, n being the total number of voltage applications bothin the case of Class 1 and Class 2. The function t_p for any value of p_e can be obtained from standard text books on statistics, but for $p_e = 95$ percent and $n = 20$, the value $t_p = 2$ may be used as an approximation:

$$U_m = \bar{U} \pm 0{\cdot}45\ s$$

Confidence limits for the value of the true standard deviation σ of the measurements may be found but with less accuracy from the expression:

$$\sigma = s\left[1 \pm \sqrt{\frac{t_p}{n-1}}\right]$$

For a more exact calculation of the confidence limits, the distribution function shall be known.

NOTE — The assumption of a Gaussian distribution leads to asymmetrical confidence limits for σ.

A-1.4 Determination of Voltages Corresponding to Very Low or Very High Disruptive Discharge Probabilities — In specific cases, a determination of very low or very high flashover probabilities of the test object is described.

Conventional assured withstand and assured flashover (*see* **4.3.2** and **4.3.3**) tests are not suitable to provide such information. The probability P of passing a conventional withstand test is:

$$P = (1 - p)^5 + 5p(1 - p)^{14}$$

where P is the probability of a flashover and account is taken of the additional 10 voltage applications if flashover occurs during the first 5.

With a test voltage corresponding to $p = 0.05$ the object will have 90 percent probability P of passing the test. With $p = 0.01$ the corresponding value is $P = 0.99$.

Thus, the fact that a test object has passed or failed such a test gives little information concerning the actual flashover probability.

Even the test method described in **A-1.2.1** is not particularly suited for the determination of low or high probabilities. A modified test procedure, therefore, described below for the particular case of low flashover probabilities. For high probabilities, *see* Note given at the end of this clause.

In a first approach, several sets of three voltage applications are made, starting at a level U_0 below the estimated withstand voltage. After every set, the voltage level is increased by a constant amount ε of between 2 percent and 5 percent of U_0. This series of tests is finished at the level U_1 when the first flashover occurs.

The test is then continued in a similar manner but with 25 voltage applications at each voltage level, starting at the level $U_2 = U_1 - 3\varepsilon$. If no flashover occurs, then the voltage level is increased by successive steps of ε until a flashover occurs. If a flashover occurs during the series of applications at U_2, then the voltage level is reduced successively by steps of 2ε until 25 applications cause no flashover. Thereafter, the level is increased by steps of ε until a flashover occurs once again.

The test result is taken as the highest voltage value which has not given flashover during any series of 25 voltage applications. This value corresponds to a flashover probability of about one percent and is below eight percent with 98 percent probability. Usually, the complete test requires about 75 voltage applications.

NOTE — The above procedure applies also to the case of high probabilities; it is sufficient to replace the words 'withstand' and 'probability' of 'withstand' by 'flashover' and 'probability of flashover'. The voltage is decreased from a high value instead of being increased from a low value.

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TO
IS : 2071 (Part II)-1974 METHODS OF HIGH VOLTAGE TESTING

PART II TEST PROCEDURES

(*First Revision*)

Alterations

(*Page* 11, *clause* 4.1.7.3) — Substitute the following for the existing clause:

'4.1.7.3 *Time to chopping* T_c — Time interval between the virtual origin and the instant of chopping.'

(*Page* 17, *clause* 4.3.4)

a) *Line* 2 — Substitute 'U_{50}' *for* 'U, 50'.

b) *Item* (b), *paragraphs* 2 *and* 3 — Substitute the following for the existing paragraphs:

'This procedure is continued, the level of each impulse being thus determined by the result of the previous one, until a sufficient number of observations has been recorded. The number of impulses n_V applied at each level U_V is then counted and the 50 percent disruptive discharge voltage is given by:

$$U_{50} = \frac{\Sigma n_v U_V}{\Sigma n_v}$$

In this equation, n_V should be $\geqslant 20$, the first level taken into account being one at which two or more impulses were applied. This avoids any appreciable error if U_k is chosen much too high or much too low. If a better accuracy in the determination of the 50 percent disruptive discharge voltage is required, the number of voltage applications should be increased but generally need not be higher than 40.'

c) *Last but one line* — Substitute the following for the existing matter:

'$\Sigma n_V \geqslant 20$'

(*Page* 21, *clause* 6.2.1, *last sentence*) — Substitute the following for the existing sentence:

'Rectangular impulse currents have virtual durations of the peak equal, within the specified tolerances, to 500 µs, 1 000 µs or 2 000 µs or between 2 000 µs and 3 200 µs.'